{EPITECH}

101PONG_

< VECTORS AND VIDEO GAMES />

101PONG

binary name: 101pong
language: everything working on "the dump"
compilation: when necessary, via Makefile, including re, clean and fclean rules

- ✓ The totality of your source files, except all useless files (binary, temp files, obj files,...), must be included in your delivery.
- ✓ All the bonus files (including a potential specific Makefile) should be in a directory named bonus.
- ✓ Error messages have to be written on the error output, and the program should then exit with the 84 error code (0 if there is no error).

Pong, an arcade game developped in 1972 by Ralph Baer (Atari), is the first ever successful video game. It was inspired by the very first video game, *Tennis for Two*, developped in 1958 by William Higinbotham on an oscilloscope.

The goal of this project is to work on a 3D version of this game (or of the *Breakout* game...). Only one paddle will be considered, located in the (Oxy) plane (which is defined by the equation $z = 0$).

Bounces on the paddle and game over will not be taken into account; in other words, **only the motion of the ball** will be considered, regardless of the context.

Your program must print:

- ✓ The velocity vector of the ball,
- ✓ The coordinates of the ball after a given amount of time,
- ✓ The angle at which the ball will hit the paddle (if it will actually hit it, at anytime from t = 0).

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Usage

```
$>  ./101pong -h
USAGE
    ./101pong x0 y0 z0 x1 y1 z1 n


DESCRIPTION
    x0  ball abscissa at time t - 1
    y0  ball ordinate at time t - 1
    z0  ball altitude at time t - 1
    x1  ball abscissa at time t
    y1  ball ordinate at time t
    z1  ball altitude at time t
    n   time shift (greater than or equal to zero, integer)
```

Suggested bonuses

- ✓ Ball acceleration management,
- ✓ A graphical interface,
- ✓ A complete 2D *Pong* game,
- ✓ A complete 2D *Breakout* game,
- ✓ A complete 3D *Pong* game,
- ✓ A complete 3D *Breakout* game,
- ✓ A spherical paddle.

Examples

Your program output has to be strictly identical to the ones below.

Terminal

```
$>  ./101pong 1 3 5 7 9 -2 4
The velocity vector of the ball is:
(6.00, 6.00, -7.00)
At time t + 4, ball coordinates will be:
(31.00, 33.00, -30.00)
The ball won't reach the paddle.
```

Terminal

```
$>  ./101pong 1.1 3 5 -7 9 2 4
The velocity vector of the ball is:
(-8.10, 6.00, -3.00)
At time t + 4, ball coordinates will be:
(-39.40, 33.00, -10.00)
The incidence angle is:
16.57 degrees
```

The incidence angle should be between 0 and 90 degrees.

Mind the float numbers precision!

v 2.5